

Gene Modulation Enabled Graph Transformer for Cancer Prognosis Prediction

Koushik Howlader¹, Md Tauhidul Islam^{2,*}, and Wei Le^{1,*}

¹Iowa State University, Iowa, USA

²Stanford University, CA, USA

*corresponding author: weile@iastate.edu & tauhid@stanford.edu

ABSTRACT

Accurate prediction of cancer progression from multi-omics data remains difficult because patient molecular profiles are highly heterogeneous. Pathway-based models incorporate prior biological knowledge and improve interpretability, but an important methodological gap remains: how to encode gene-to-pathway representations that capture patient-specific genomic variation while preserving stable gene identity. Here, we present PATH, a pathway-aware graph transformer for accurate and interpretable cancer progression prediction. PATH learns patient-conditioned gene embeddings using Feature-wise Linear Modulation (FiLM) technique, allowing each gene representation to retain its biological identity while adapting to an individual patient's CNV and mutation profile. These gene embeddings are aggregated into pathway tokens and processed by a graph transformer with a soft-mask mechanism that refines known pathway interactions and permits candidate new pathway connections to emerge. PATH achieves an F1 score of 0.8766 on pan-cancer metastasis prediction, representing an 8.8% improvement over the strongest baseline. Beyond predictive performance, PATH recovers biologically interpretable signals at the gene, pathway, and pathway-pathway interaction levels. These results show that patient-specific gene representation learning, combined with biologically guided pathway-interaction modelling, provides a principled strategy for improving both accuracy and interpretability in cancer progression prediction.

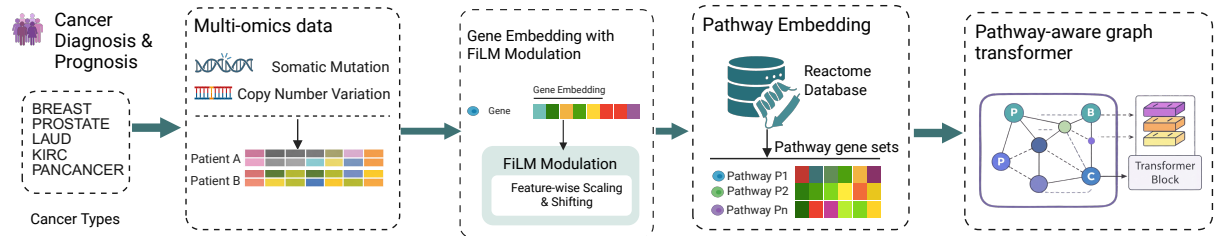
1 Introduction

Cancer progression is shaped by diverse combinations of genomic alterations, and patients with the same clinical label can carry distinct molecular profiles^{1,2,3}. This molecular heterogeneity makes progression prediction difficult. Deep learning methods have improved cancer classification and prognosis modelling from genomic and multi-omics data^{4,5,6,7}, but strong prediction alone is not sufficient for cancer applications. A useful model should also indicate which genes, pathways, and pathway interactions contribute to the prediction.

Biologically informed models address this need by incorporating prior knowledge such as gene-pathway memberships and pathway databases^{8,9,10,11}. Pathway-based neural networks such as PathCNN and P-NET use biological hierarchies to improve interpretability¹². Graph-based multi-omics models such as MOGONET and MOGAT use graph learning to integrate heterogeneous molecular information¹³. Transformer-based pathway models have also been proposed; for example, Pathformer uses a pathway-informed transformer framework for disease diagnosis and prognosis using multi-omics data¹⁴. Despite these advances, a key challenge remains: gene-level alterations must be converted into pathway-level representations in a way that preserves both stable gene identity and patient-specific molecular context.

Here, we introduce **PATH**, a pathway-aware Graph Transformer for cancer progression prediction. PATH first learns a base embedding for each gene and then conditions this embedding on the patient's CNV and mutation profile using Feature-wise Linear Modulation (FiLM)¹⁵. This design allows the same gene to have different representations in different patients while retaining a shared gene identity across the cohort. The patient-conditioned gene embeddings are then aggregated into pathway representations. PATH next uses a Graph Transformer to model interactions among pathways, with Reactome-derived pathway relationships providing the biological graph prior¹⁶. A soft-mask mechanism penalizes, but does not fully block, attention between pathway pairs that are not connected in Reactome, allowing the model to refine known pathway relationships and identify candidate non-Reactome interactions.

a. Overview of the PATH Framework



b. Pathway-aware graph transformer

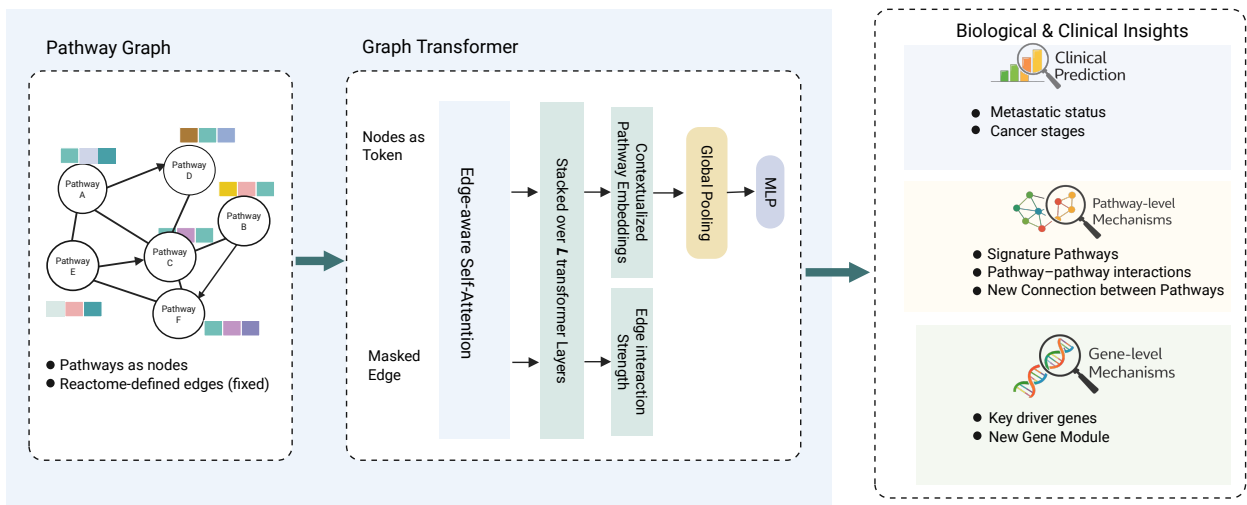


Figure 1. Overview of PATH: a graph transformer for modelling pathway–pathway interactions. (a) For each cancer cohort, patient multi-omics profiles (somatic mutation and copy-number variation) are used as input. PATH performs *gene-level representation learning* to produce patient-specific gene features, aggregates these features into *pathway embeddings* using Reactome gene sets, and generates pathway tokens summarising pathway activity. The pathway tokens are passed to a *pathway-aware graph transformer* to model pathway crosstalk and produce predictions and interpretable outputs. (b) **Graph transformer to model pathway–pathway interaction.** Pathways are treated as nodes in a graph, with Reactome-defined edges providing a biological prior. The transformer performs contextualised, edge-aware self-attention over pathway tokens; soft-masked edges improve robustness to incomplete pathway knowledge while learning data-driven interaction strengths. Stacking L layers yields contextualised pathway embeddings, which are pooled and passed to an MLP for clinical prediction. Learned attention and interaction strengths enable interpretability, including signature pathways, pathway–pathway interactions, model-inferred connections, and key driver genes.

We evaluate PATH on multiple cancer progression tasks, including the three main cohorts reported here and two additional datasets described in the Supplementary Material. Across these settings, PATH outperforms the compared baseline models in the benchmarking analysis (Fig. 2), and ablation analyses show that both FiLM-based patient conditioning and Graph Transformer modelling contribute to performance (Fig. 3). Beyond classification, PATH identifies interpretable signals at multiple biological scales, including COSMIC²- or literature-supported genes, pathway programmes, and pathway–pathway interaction shifts that can be treated as computational hypotheses. These results show that patient-conditioned gene representation learning and biologically guided pathway interaction modelling can improve cancer progression prediction while preserving interpretability.

2 Results

2.1 Overview of PATH

A key challenge in cancer progression prediction is the high heterogeneity of patient molecular profiles, as the same cancer type can arise from different combinations of molecular alterations. For pathway-based models, an additional challenge is how to construct pathway representations from genes. PATH (Fig. 1) addresses these challenges through two main components: patient-conditioned gene embedding using Feature-wise Linear Modulation (FiLM)², and pathway interaction learning using a graph transformer.

First, PATH learns a base embedding for each gene. This base embedding represents the biological identity of the gene. PATH then adjusts this embedding using FiLM, a conditioning method that learns how to scale and shift one representation using another input. In PATH, this conditioning input is the patient's CNV and somatic mutation profile. This means the same gene can have different representations in different patients depending on their molecular alterations. To the best of our knowledge, PATH is the first pathway-based cancer progression model to use FiLM-style modulation to construct patient-conditioned gene embeddings from genomic alterations. As a result, PATH produces gene representations that reflect both the gene's biological identity and the patient's individual molecular state.

PATH processes each patient's genomic data through four stages to generate a cancer progression prediction (Fig. 1b). In the first stage, each gene receives a base embedding, which is then conditioned by the patient's mutation and CNV profile using FiLM. In the second stage, the gene representations within each biological pathway are pooled into a single representation. In the third stage, PATH uses a pathway-aware graph transformer to model interactions among pathways. Each pathway is treated as a node in the graph, and known Reactome pathway relationships² are used to initialize edges. Learning through graph transformers allows each pathway to update its representation by attending to other related pathways, rather than treating pathways as independent features. The graph transformer is central to PATH because it potentially captures pathway–pathway interactions important for the prediction and learns how signaling relationships change during cancer progression.

The known biological connections are encoded as a pathway–pathway adjacency matrix derived from the *Reactome* database. To allow the model to detect connections beyond *Reactome*, PATH applies a soft-masking strategy. This strategy penalizes, but does not fully block, communication between pathway pairs that are not connected in *Reactome*. Together, the graph transformer and soft mask allow PATH to capture both established and potentially novel pathway interactions. In the fourth stage, the pathway representations are combined into a single patient-level representation using attention-based pooling. This final representation is passed to a classifier to predict cancer progression.

At inference, the trained model takes a new patient's mutation and CNV profiles as input and returns a class prediction. Beyond classification, PATH also supports model interpretation at multiple scales. First, marker genes and marker pathways are identified using SHAP² analysis. Then, pathway–pathway attention weights are extracted for each disease state to identify pathway interactions during cancer progression.

2.2 Systematic benchmarking of PATH across diverse cancer progression tasks

To assess generalization across cancer types, disease contexts, and prediction tasks, we evaluate PATH on three independent cohorts. PANCANCER ($n = 8,893$; 8,532 primary, 361 metastatic)² combines samples from 33 cancer types, covering a wide range of tumour origins, and tests whether PATH performs consistently across a highly varied dataset. PRAD ($n = 1,011$; 333 primary, 678 metastatic)² focuses on prostate cancer metastatic progression, a specific cancer type with well-studied biology, enabling direct biological validation of model predictions. BLCA ($n = 434$; 138 early stage, 296 late stage)² addresses early versus late-stage bladder cancer classification, testing whether PATH generalizes to a different type of prediction task. Together, the three cohorts span variation in cancer type, disease context, and prediction task—the conditions under which a cancer prediction model must prove reliable.

To benchmark PATH within this landscape, we selected seven state-of-the-art baselines spanning three major architectural families in multi-omics cancer prediction. The first family comprises pathway-based models: PathCNN², P-NET², and Pathformer², which map genes onto biological pathways. The second family comprises patient-graph models: MOGONET² and MOGAT², which connect patients with similar omics profiles and classify them using graph neural networks. The third family comprises feature-fusion models: eiCNN and liCNN², which apply convolutional neural networks to multi-omics

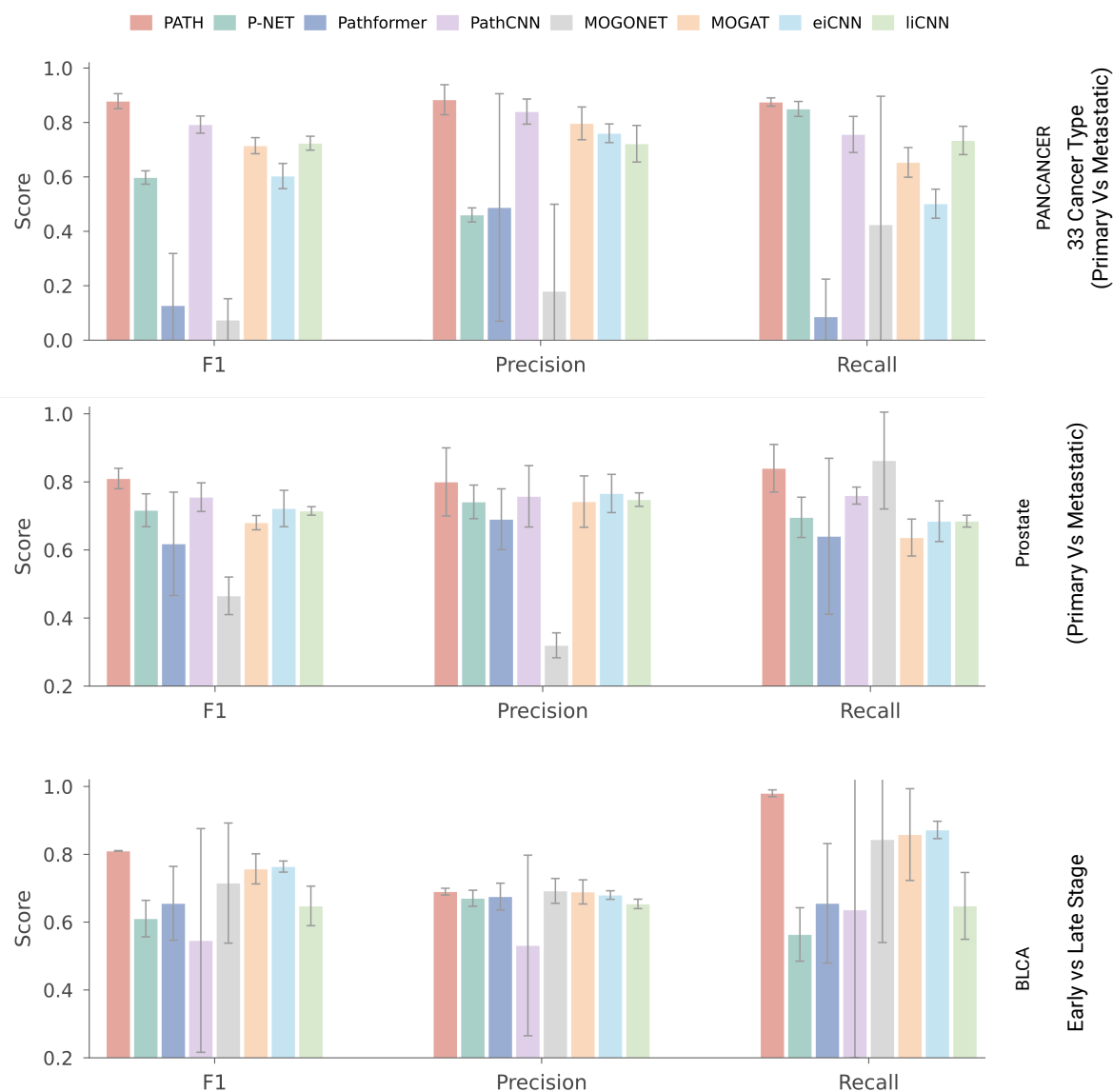


Figure 2. Comparative performance of PATH and baseline models across all three cancer cohorts. Bar plots show mean $\pm$ standard deviation of F1 score, precision, and recall across five-fold cross-validation. **(a)** PANCANCER (33 cancer types, primary vs. metastatic). **(b)** PRAD (prostate cancer, primary vs. metastatic). **(c)** BLCA (bladder cancer, early vs. late stage). Baselines include PathCNN, P-NET, eiCNN, liCNN, MOGONET, and MOGAT.

features integrated either before network processing (early fusion) or after modality-specific processing (late fusion). All seven baselines were evaluated under the same five-fold cross-validation setup, with performance reported as the mean across folds and assessed using F1 score, precision, and recall. Against these baselines, PATH achieves the best performance across all three cohorts (Fig. 2). Specifically, on PANCANCER, PATH achieves F1 0.88 ± 0.03 (precision 0.89 ± 0.06 , recall 0.86 ± 0.04) (see Fig. 2a), which was 10.9% higher than the second-best model, PathCNN. Similarly, in P1000 (prostate cancer), PATH achieved an F1 score of 0.81 ± 0.03 , representing a 7.3% improvement over the next-best model, PathCNN (see Fig. 2b). Finally, on BLCA, PATH achieved an F1 score of 0.810 ± 0.001 , a 6.0% improvement over the next-best model, eiCNN, together with a near-perfect recall of 0.98 ± 0.02 (see Fig. 2c).

To determine the contribution of PATH's main components, we performed ablation experiments on the PANCANCER dataset (Fig. 3), with corresponding results for PRAD and BLCA provided in Supplementary Fig. X. First, replacing the Vanilla Transformer with the pathway-aware Graph Transformer increased F1 from 0.66 to 0.73, precision from 0.73 to 0.76, and recall

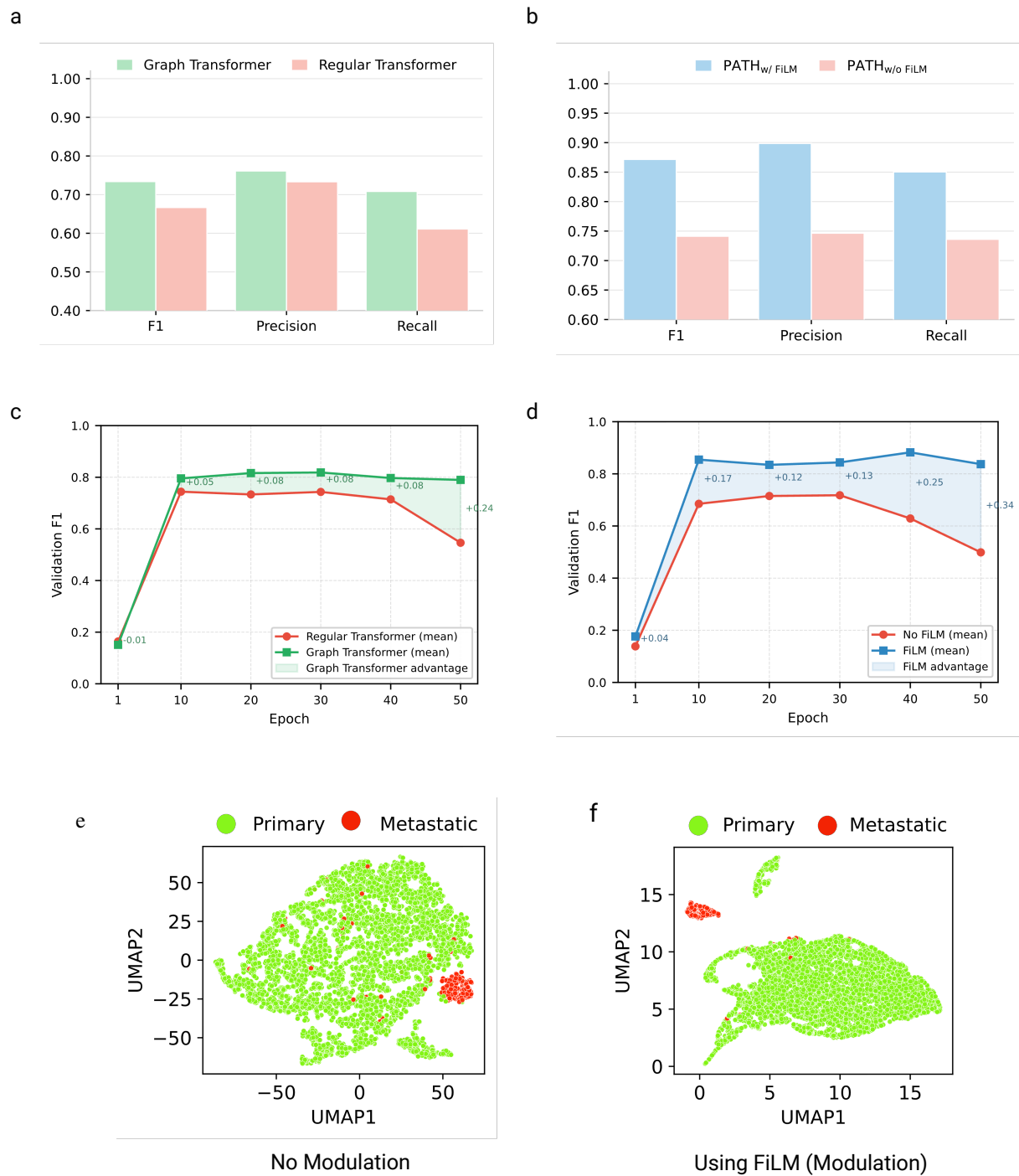


Figure 3. Ablation analysis of Graph Transformer modelling and FiLM-based patient conditioning in PATH. (a) Test-set F1, precision, and recall for the pathway-aware Graph Transformer compared with a regular Transformer. (b) Test-set F1, precision, and recall for PATH with FiLM compared with PATH without FiLM. (c) Validation-F1 trajectories for the Graph Transformer and regular Transformer across training epochs. (d) Validation-F1 trajectories for FiLM and no-FiLM models across training epochs. (e,f) UMAP projections of learned representations without FiLM conditioning (e) and with FiLM conditioning (f).

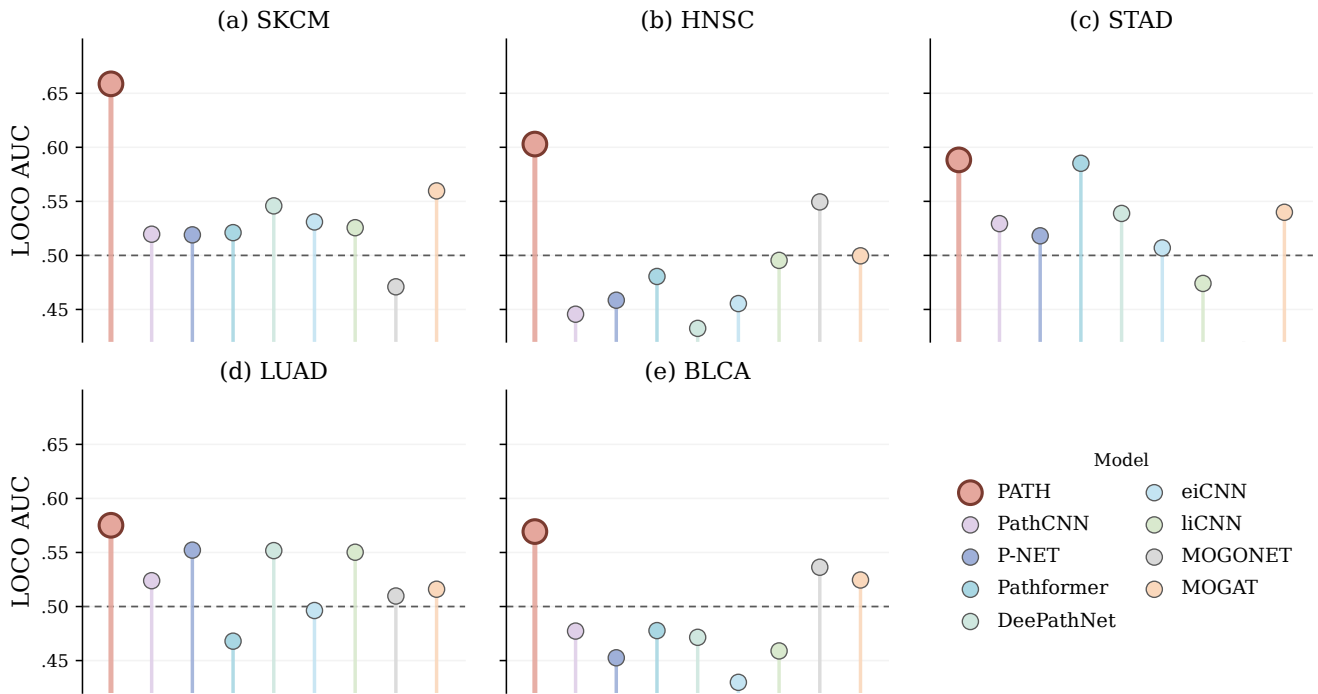


Figure 4. PATH generalizes to unseen cancer types under leave-one-cohort-out (LOCO) evaluation on CNV + mutation. Each panel is a held-out cancer cohort: every model is trained on the pooled remaining cohorts and evaluated on the entire held-out cohort (early- vs late-stage classification). Dots show test AUC for PATH and eight baselines (same colour key); PATH’s marker is enlarged and outlined. The dashed line marks chance (AUC = 0.5). Panels show the five cohorts in which PATH attains the highest AUC of all nine models **(a) SKCM**, **(b) HNSC**, **(c) STAD**, **(d) LUAD** and **(e) BLCA**; across all eight LOCO cohorts PATH ranks first in five and top-two in seven, and is the only model whose AUC stays above chance in every held-out cohort.

from 0.61 to 0.71 (Fig. 3a). Second, when the Graph Transformer was held fixed, adding FiLM increased F1 from 0.74 to 0.87, precision from 0.74 to 0.90, and recall from 0.73 to 0.85 (Fig. 3b). The validation curves further show that the Graph Transformer consistently outperformed the Vanilla Transformer during training (Fig. 3c), while FiLM produced a larger and more stable improvement over the non-FiLM model (Fig. 3d). Consistent with these quantitative results, the UMAP projections show clearer separation between primary and metastatic samples when FiLM conditioning is used (Fig. 3e,f). Together, these results demonstrate that both pathway-aware interaction modeling and patient-specific FiLM conditioning contribute to PATH’s performance.

2.3 PATH generalizes to unseen cancer types under leave-one-cohort-out evaluation

Within-cohort benchmarking shows that PATH is accurate on each cancer individually, but a more demanding test is whether it transfers to a cancer type it has never seen. We therefore evaluated early- vs late-stage prediction under a leave-one-cohort-out (LOCO) protocol on the CNV + mutation modality: for each of eight TCGA cohorts, every model is trained on the pooled remaining seven cohorts and tested on the entire held-out cohort, so the test cancer type is absent from training. Under this cross-cancer transfer setting PATH attains the highest AUC of all nine models in five of the eight held-out cohorts—SKCM (0.659), HNSC (0.603), STAD (0.588), LUAD (0.575) and BLCA (0.569) (Fig. 4)—and ranks in the top two in seven of eight. Notably, PATH is the only model whose AUC stays above chance in every held-out cohort (median AUC 0.58), whereas the flat and pathway baselines fluctuate around chance (medians 0.48–0.52; Supplementary Table 6). These results indicate that PATH’s pathway-guided inductive bias aids generalization across cancer types from sparse copy-number and mutation signal, where architectures without an explicit biological prior tend to overfit the training cohorts.

2.4 PATH reveals marker genes and pathways

Strong accuracy tells us that PATH works, but not *why* it works, so we next asked what the model actually relies on. To make its decisions interpretable, we applied SHAP attribution to the trained PANCANCER model, which spans 33 cancer types and assigns every gene and pathway a signed contribution to each prediction. Because a genuine metastasis marker should be a feature the model weights more heavily for metastatic than for primary tumours, we scored each gene by the increase in its

mean SHAP value (mutation and CNV combined) from primary to metastatic samples, and kept the ten strongest (Fig. 5a; Table 11).

A marker list is only convincing if it recovers known biology, so we cross-checked these ten genes against COSMIC and the literature. Reassuringly, eight are established cancer genes: four—*BRAF*, *TP53*, *MITF*, and *EP300*—are backed by both COSMIC and the literature as tumour-progression genes^{2,3,22,23}, and four more—*GHR*, *ADAMTS18*, *XDH*, and *MUC16*—have experimental or clinical links to metastasis^{2,3,22,23}. The remaining two, *SCN11A* and *EEF1E1*, carry cancer evidence but no established metastatic role, so we flag them as hypotheses for validation rather than firm claims^{2,3,23}.

Genes exert their effects through pathways, so we then repeated the same attribution analysis at the pathway level (Fig. 5b; Table 12). Splitting each pathway's attribution into its primary and metastatic parts confirmed that the top pathways all push toward metastasis, and that they map onto well-known progression machinery—FGF/FGFR signaling and the RAS–RAF–MEK–ERK cascade^{2,3,22,23}.

A pan-cancer claim, however, could still be driven by a single dominant cohort, so we next checked whether the markers recur across cancers individually. Indeed, many genes and pathways rank near the top in three or more cohorts (Fig. 5c,d), showing that the signal generalizes rather than reflecting one tumour type; a few genes (*HRAS*, *SRC*, *FGF5*, *RAC1*) recur across the shared pathways. Finally, to test whether these markers form a coherent picture rather than a loose list, we clustered the marker pathways by shared genes. They separated into three interpretable modules—gap junction, NMDA-receptor/Ca²⁺–RAS, and a large FGFR/EGFR/ERBB receptor-tyrosine-kinase module (Fig. 5e)—indicating that PATH's markers are organized signaling programs, not scattered features.

2.5 PATH uncovers metastasis-associated pathway interactions and hubs

So far we have looked at genes and pathways one at a time. PATH's defining feature, however, is that it also learns how pathways interact, so we next turned to the interaction network itself and asked what changes as a tumour becomes metastatic (Fig. 6).

First, we compared the pathway–pathway attention, or “crosstalk,” between primary and metastatic tumours. Most of the largest changes were interactions that grew stronger in metastasis, and those that survived multiple-testing correction are outlined in black (Benjamini–Hochberg FDR < 0.05; Fig. 6a). The biggest gains centred on FGF/FGFR–Spry signalling^{2,24}, L1CAM cell adhesion^{2,25}, and collagen/extracellular-matrix pathways—the same programmes that dominated the marker pathways above, and all established contributors to tumour invasion and metastasis.

Second, because the soft mask can also connect pathways that Reactome leaves unlinked, PATH can propose entirely new interactions. Three of these novel links were statistically significant (coral edges in Fig. 6b): extracellular matrix organization with the circadian clock—in line with emerging evidence that circadian disruption contributes to cancer²⁶—and both cell–cell and cell junction organization with cilium assembly. We stress that these are class-dependent statistical associations rather than proven physical interactions, so we report them as hypotheses for follow-up.

Third, these edge-level changes add up to a global shift. When each pathway is placed by its attention pattern, the whole map moves systematically from the primary to the metastatic state (Fig. 6c), showing that metastasis rewires the interaction network broadly rather than at a few isolated edges.

Finally, we asked which pathways organize this network. We scored each pathway as a hub, giving it a high value when the pathway both carries strong metastatic attribution and connects to other high-attribution pathways (Methods, Eqs. 31–32). *Sprouty regulation of FGF signaling* emerged as the clear master hub (Fig. 6d): it links the FGFR1–4 negative-regulation pathways to oncogenic MAPK/RAS signaling. This placement matches known biology, in which Sprouty is a feedback regulator of FGF/FGFR signalling that tunes the downstream RAS/MAPK cascade²⁷. Taken together, these results show that PATH captures metastatic progression not only in individual genes and pathways but in the way pathways interact—strengthening known crosstalk, proposing new links, and converging on an FGF/FGFR–MAPK hub.

2.6 PATH predicts patient survival and highlights aggressiveness-linked pathways

So far we have used PATH for classification. A more demanding question is whether the same model can estimate *how long* a patient will survive, so we next turned to survival prediction. These survival cohorts (LGG, LUSC and UVM) are separate from the classification datasets used above; here each patient is labelled by overall survival time rather than by tumour stage or metastatic status, and the task is to rank patients by risk rather than to assign a class. Using only copy-number and somatic-mutation profiles, we trained PATH to predict overall survival and scored it with the concordance index (C-index)²⁸, where 0.5 denotes random risk ranking and 1.0 a perfect one. As before, scores were computed within each of five cross-validation folds and then averaged, and PATH was compared with six pathway- and multi-omics baselines (Fig. 7).

Across three cohorts, PATH reached the highest concordance among the compared models: LGG (C-index 0.77; Fig. 7a), LUSC (0.54; Fig. 7b) and UVM (0.66; Fig. 7c). A high C-index, however, is only meaningful if it separates patients whose outcomes truly differ, so we next divided the LGG patients into high- and low-risk groups by their PATH risk score. The two groups showed markedly different outcomes, with high-risk patients dying far sooner (Kaplan–Meier curves, log-rank

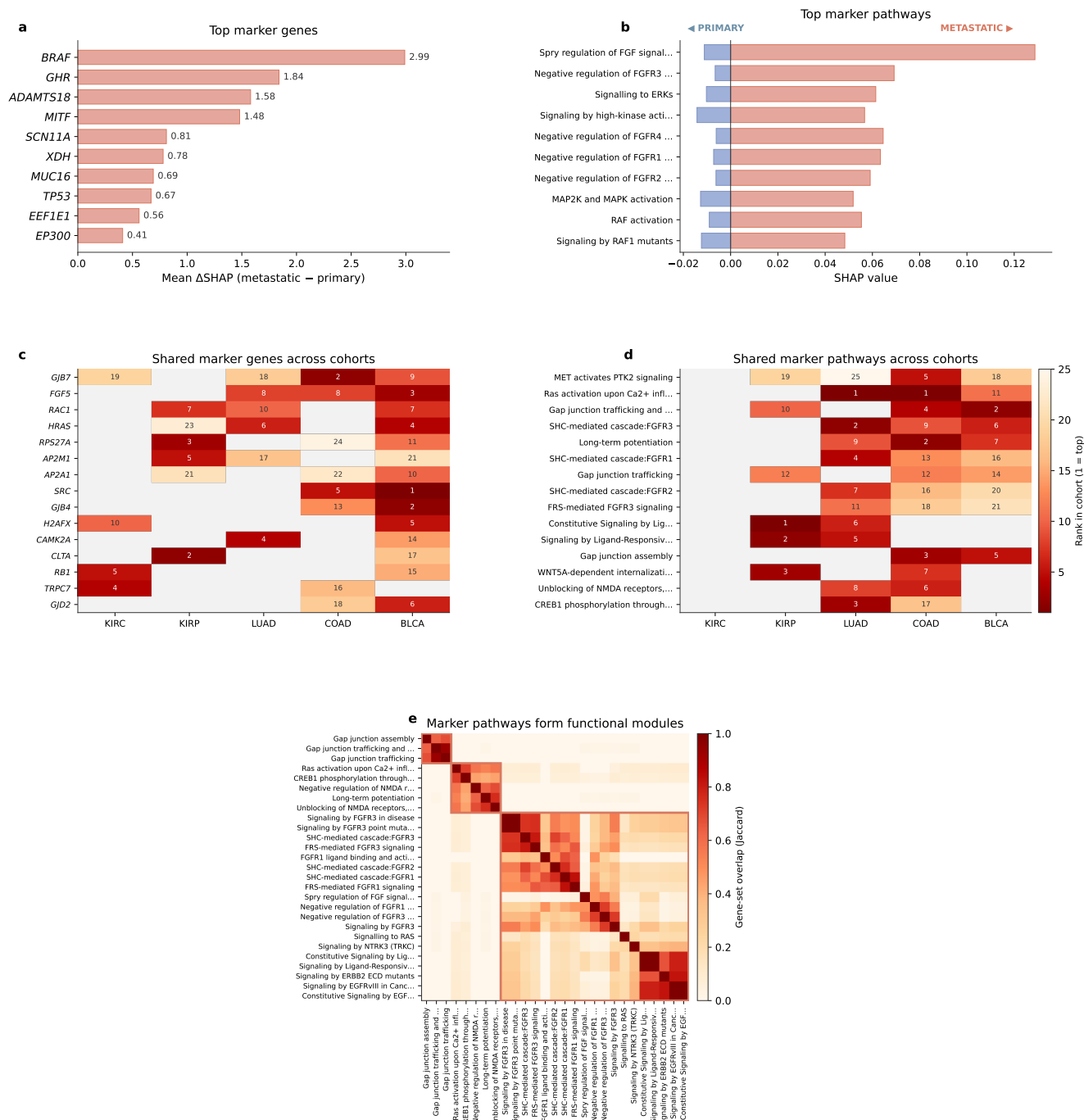


Figure 5. PATH reveals marker genes and pathways of metastatic progression. (a) Top marker genes ranked by the mean shift in attribution between metastatic and primary samples (Δ SHAP = $\text{mean}_{\text{met}} - \text{mean}_{\text{prim}}$); larger values indicate stronger association with the metastatic state. (b) Top marker pathways shown as a primary-versus-metastatic decomposition of the SHAP attribution; coral bars (right) denote metastatic-increasing contribution and blue bars (left) denote primary-increasing contribution. (c) Cross-cohort recurrence of the shared marker genes: each cell gives the within-cohort SHAP rank (1 = strongest) of a gene across five cancer cohorts; grey cells indicate the gene is not among that cohort's top markers. (d) Cross-cohort recurrence of the shared marker pathways, displayed as in (c). (e) Marker pathways organize into functional modules. Pairwise gene-set (Jaccard) overlap among the marker pathways, hierarchically clustered; coral boxes delineate the recovered modules (a gap-junction module, an NMDA-receptor/ Ca^{2+} -RAS module, and a large FGFR/EGFR/ERBB receptor-tyrosine-kinase module).

$p = 6.7 \times 10^{-11}$; Fig. 7d). The same pattern held at the individual level: patients who died carried higher PATH risk scores than those who were still alive (Fig. 7e).

Having established that PATH ranks risk well, we then asked *which* pathways it relies on when it flags an aggressive tumour. Within each fold we compared pathway attention between aggressive (high-risk) and indolent (low-risk) LGG patients. The volcano plot summarizes this comparison: every point is a pathway, its horizontal position gives how far attention shifted between the two groups and its vertical position how significant that shift is, coloured by whether the shift pointed in the same direction across all five folds (Fig. 7f). Reassuringly, the most strongly shifted pathways moved consistently, indicating a reproducible attention signal rather than fold-specific noise.

Focusing on the pathways that gained attention in aggressive tumours in every fold (Fig. 7g), we found a coherent set of stress- and progression-related programmes: TP53-driven DNA-damage and cell-cycle checkpoints², TGF- β /SMAD2–SMAD3–SMAD4 transcriptional signalling², and the KEAP1–NFE2L2/NFE2L2 oxidative-stress response², alongside MITF-M lineage and interferon- α/β programmes—all well-established cancer-related processes, which lends biological plausibility to what the model has learned.

These pathways, however, describe what the model *looks at* (association), not proven causes. To make this explicit, we tested the single top-ranked pathway on its own using its raw copy-number activity, and it did not separate survival ($p = 0.70$). PATH's accuracy therefore comes from combining many pathways rather than from any one of them. Taken together, these results show that PATH extends beyond classification to survival prediction, delivering competitive risk scores together with biologically sensible pathway attention.

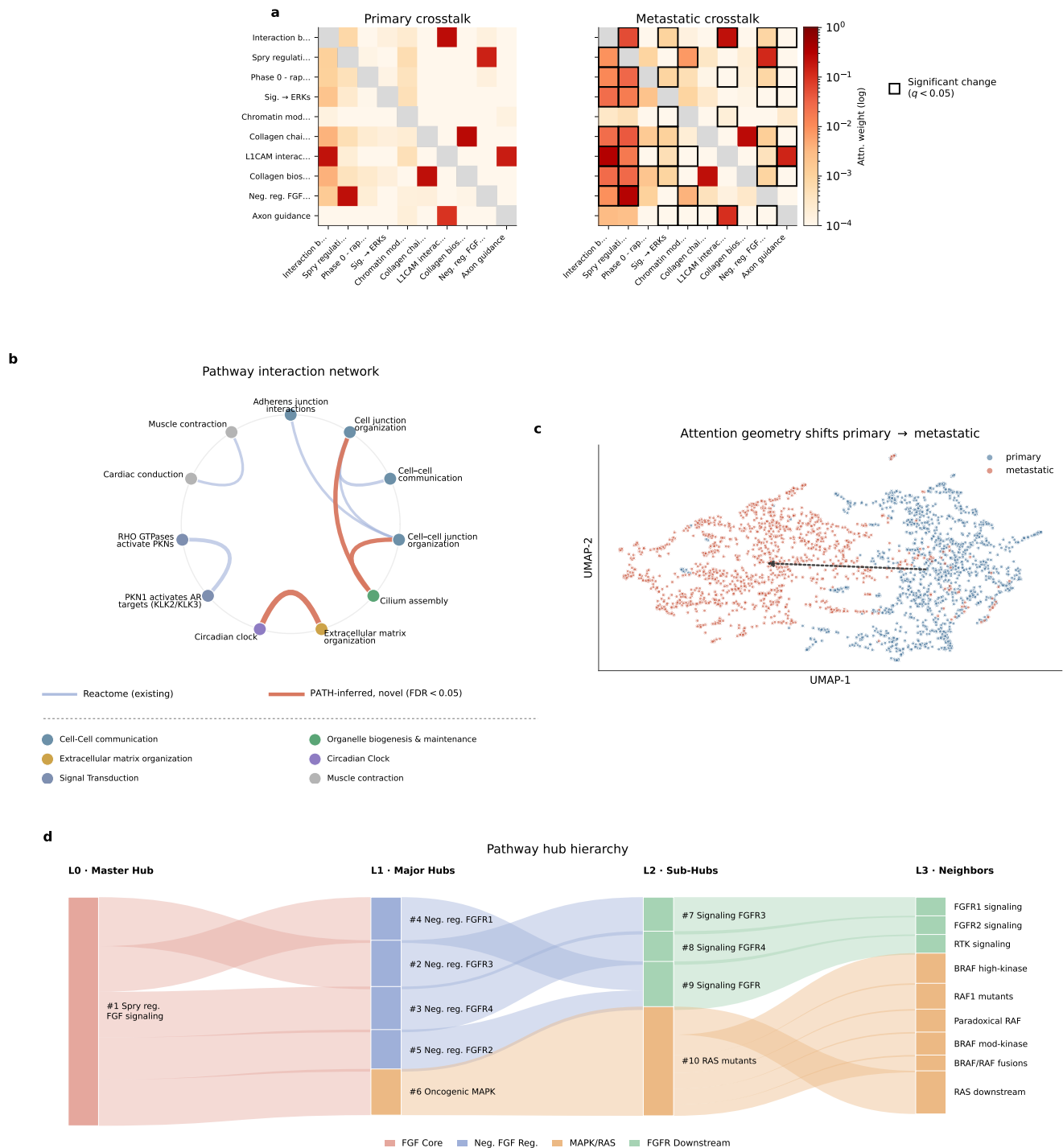


Figure 6. PATH uncovers metastasis-associated pathway interactions and hubs. (a) Pathway–pathway attention (crosstalk) among the top interacting pathways in primary (left) and metastatic (right) tumours, on a shared log scale; solid black borders on the metastatic panel mark interactions whose change between metastatic and primary samples is statistically significant (Benjamini–Hochberg FDR < 0.05). (b) Pathway interaction network: blue arcs are curated Reactome interactions and coral arcs are model-inferred novel interactions absent from the Reactome prior that pass FDR < 0.05 (extracellular matrix organization–circadian clock, and cell–cell/cell junction organization–cilium assembly). (c) UMAP of pathway attention embeddings; arrows trace the shift of each pathway from the primary (blue) to the metastatic (coral) state. (d) Hierarchical hub organization of the metastatic interaction network. Combining positive metastatic attribution with the Reactome network identifies *Spry regulation of FGF signaling* as the master hub (L0), which connects through the FGFR1–4 negative-regulation major hubs (L1) and FGFR/RAS sub-hubs (L2) to their downstream neighbour pathways (L3); colours denote functional groups.

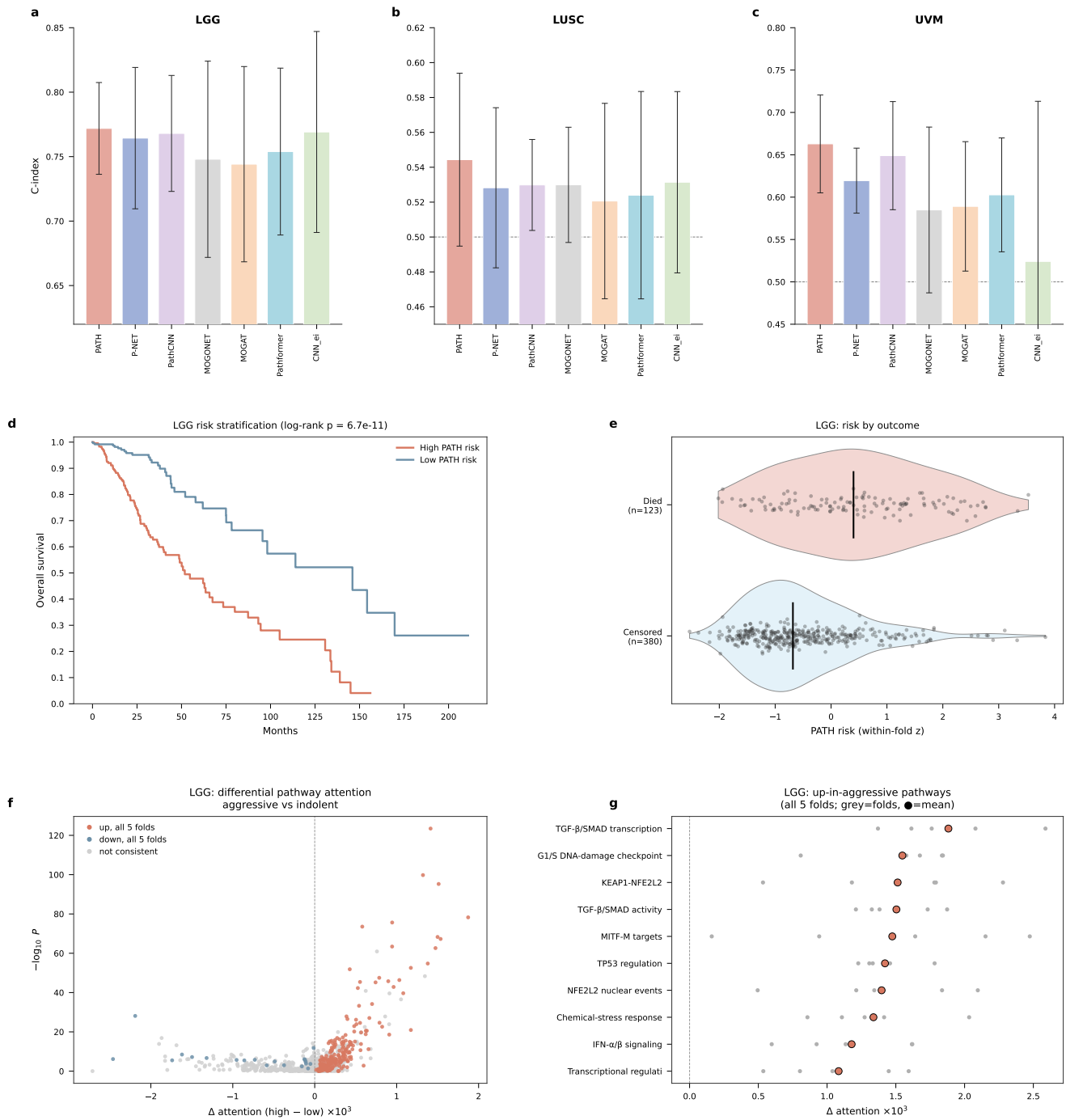


Figure 7. PATH gives competitive survival prediction and highlights aggressiveness-linked pathways. (a–c) C-index for PATH versus six baselines in (a) LGG, (b) LUSC and (c) UVM (mean $\pm$ s.d.; dashed line, 0.5). (d) LGG survival curves for high- versus low-PATH-risk patients (log-rank $p = 6.7 \times 10^{-11}$). (e) PATH computed risks in LGG patients who died versus censored. (f) Differential pathway-attention volcano for LGG (aggressive versus indolent), coloured by five-fold direction consistency. (g) Top pathways with higher attention in aggressive LGG across all five folds (grey, per-fold; filled, mean).

3 Discussions

We built PATH, a pathway-aware graph transformer for cancer prediction. PATH does two things that earlier pathway models do not. It gives each gene a stable identity and then reshapes that identity to fit each patient’s own mutation and CNV profile using FiLM. It also treats pathways as connected nodes and lets them share information through Reactome-guided attention, instead of scoring each pathway on its own.

These two ideas worked across very different tasks. PATH reached the best F1 among all compared methods on pan-cancer metastasis and on prostate metastasis, and it also separated early from late bladder cancer well. We then moved to a much harder task: predicting how long a patient survives from CNV and mutation alone. Here PATH stayed competitive, reaching the highest concordance among the compared baselines in the LGG, LUSC, and UVM cohorts. So the same model handles both yes/no progression labels and continuous survival time.

Good accuracy is only useful if the risk scores mean something. In LGG, splitting patients by PATH risk cleanly separated their survival curves (log-rank $p = 1.7 \times 10^{-10}$), and patients who died carried higher risk than those still alive. PATH is therefore not ranking patients by chance; it captures a real signal of how aggressive a tumour is.

PATH also explains itself. For metastasis it pointed to well-known cancer genes and pathways: FGF/FGFR feedback and MAPK signalling in the pan-cancer cohort, and AR, RUNX2, and mTORC1 programmes in prostate cancer. For survival, we asked which pathways the model attends to most when it flags an aggressive tumour. Across all five folds it consistently highlighted DNA-damage checkpoint, TGF- β /SMAD, RUNX3, and oxidative-stress (KEAP1–NFE2L2) programmes, all of which are known to drive cancer. These pathways show what the model looks at, not proven causes: when we tested the top pathway on its own using raw copy-number activity, it did not separate survival ($p = 0.70$). PATH’s strength therefore comes from combining many pathways, not from any single one.

Beyond single pathways, PATH can also read how pathways act together. In the pan-cancer analysis it found stronger co-activation between adhesion pathways and between FGFR feedback and Sprouty/FGF signalling in metastatic tumours, matching known invasion and FGFR–RAS–ERK biology^{2,3,7}. It also proposed a few pathway links that are absent from Reactome, such as an extracellular-matrix–circadian-clock connection^{2,7}. We treat these as testable ideas, not facts.

PATH has clear limits. It uses only mutation and CNV, so adding gene expression, methylation, or protein data may help capture states these two modalities miss. It depends on Reactome, so pathways missing from that database cannot be modelled directly. Our survival results are competitive but not the single best across every method, and all findings come from retrospective TCGA cohorts, so prospective validation in independent clinical cohorts is still needed. Unsupported genes, weaker pathway signals, and non-Reactome links remain predictions that must be checked experimentally.

In short, PATH shows that learning patient-specific gene-to-pathway representations, together with pathway-to-pathway interaction modelling, can predict both cancer progression and patient survival while pointing to biology that makes sense. By showing how pathway activity and coordination shift between disease states, PATH moves from static pathway lists toward a dynamic picture of cancer^{2,7}.

4 Methods

4.1 Cohorts and input representation

PATH was evaluated on three main cancer progression tasks. The pan-cancer cohort was assembled from TCGA samples spanning 33 tumour types and was used for primary-versus-metastatic classification^{2,7}. The prostate cancer cohort was used for disease-specific primary-versus-metastatic classification². The TCGA BLCA cohort was used for early-versus-late stage classification, where stage I–II samples were assigned to the early-stage class and stage III–IV samples to the late-stage class². Additional datasets used for supplementary analyses were processed with the same workflow.

For each patient i and gene g , somatic mutation status was encoded as a binary variable $m_{ig} \in \{0, 1\}$, and copy-number variation (CNV) was represented as a continuous value c_{ig} . Mutation and CNV matrices were aligned to the same patient and gene order. CNV values were standardized within each cross-validation fold using training-set statistics only,

$$\tilde{c}_{ig} = \frac{c_{ig} - \mu_g^{\text{train}}}{\sigma_g^{\text{train}} + \varepsilon}, \quad (1)$$

where $\varepsilon = 10^{-8}$ was used for numerical stability. The input for a patient-gene pair was therefore

$$\mathbf{x}_{ig} = [m_{ig}, \tilde{c}_{ig}]^\top. \quad (2)$$

Pathway definitions and pathway membership were obtained from Reactome². Each pathway p was represented by a gene set $\mathcal{G}_p$, after retaining genes present in the processed mutation and CNV matrices. The pathway graph prior was defined by

an adjacency matrix $\mathbf{A} \in \mathbb{R}^{P \times P}$ over P pathways. In the sparse-graph setting, $\mathbf{A}$ was loaded from the pathway adjacency file, symmetrized, and normalized,

$$\mathbf{A} \leftarrow \frac{(\mathbf{A} + \mathbf{A}^\top)/2}{\max(\mathbf{A}) + \varepsilon}. \quad (3)$$

For ablation, a full-graph setting replaced this prior with a complete pathway graph, allowing all pathway pairs to attend to each other.

4.2 PATH architecture

PATH maps patient-level genomic profiles to class probabilities through four stages: patient-conditioned gene embedding, pathway token construction, graph-transformer-based pathway interaction modelling, and attention-based patient readout.

Patient-conditioned gene embedding

Each gene g has a learned base embedding $\mathbf{e}_g \in \mathbb{R}^d$, where $d = 64$ in the default model. To make this gene representation patient-specific, PATH uses Feature-wise Linear Modulation (FiLM)[?]. A two-layer multilayer perceptron maps the mutation and CNV input $\mathbf{x}_{ig}$ to scale and shift parameters,

$$[\gamma_{ig}^{\text{raw}}, \beta_{ig}] = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{x}_{ig} + \mathbf{b}_1) + \mathbf{b}_2. \quad (4)$$

The scale parameter is constrained to be positive,

$$\gamma_{ig} = \text{softplus}(\gamma_{ig}^{\text{raw}}) + 10^{-3}, \quad (5)$$

and the patient-conditioned gene embedding is

$$\mathbf{h}_{ig} = \gamma_{ig} \odot \mathbf{e}_g + \beta_{ig}, \quad (6)$$

where $\odot$ denotes element-wise multiplication. The final FiLM layer was initialized to produce near-identity modulation at the start of training, with zero output weights, zero shift bias, and normally initialized gene embeddings.

Pathway token construction

For each pathway p , PATH aggregates the conditioned gene embeddings $\{\mathbf{h}_{ig} : g \in \mathcal{G}_p\}$ into a pathway token. Because pathways contain different numbers of genes and not all member genes are equally informative, aggregation is performed using learned attention rather than uniform averaging. For gene g in pathway p ,

$$s_{ipg} = \mathbf{u}^\top \tanh(\mathbf{W} \mathbf{h}_{ig} + \mathbf{b}_p), \quad (7)$$

where $\mathbf{b}_p$ is a pathway-specific bias. Attention weights are normalized over pathway members,

$$\alpha_{ipg} = \frac{\exp(s_{ipg})}{\sum_{g' \in \mathcal{G}_p} \exp(s_{ipg'})}, \quad (8)$$

and the pathway token is

$$\mathbf{z}_{ip} = \sum_{g \in \mathcal{G}_p} \alpha_{ipg} \mathbf{h}_{ig}. \quad (9)$$

This yields a pathway-token matrix $\mathbf{Z}_i \in \mathbb{R}^{P \times d}$ for patient i .

Laplacian pathway positional encoding

To encode the position of each pathway in the prior graph, PATH computes a normalized graph Laplacian,

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}, \quad (10)$$

where $\mathbf{D}$ is the pathway-degree matrix. The first $k = 16$ eigenvectors of $\mathbf{L}$ were used as Laplacian positional features[?]. These features were concatenated with normalized node degree and projected to the model dimension,

$$\mathbf{p}_p = \mathbf{W}_{\text{PE}} \left[\mathbf{v}_p^{(1)}, \dots, \mathbf{v}_p^{(k)}, \frac{d_p}{P + \varepsilon} \right] + \mathbf{b}_{\text{PE}}. \quad (11)$$

During training, random sign flips were applied to eigenvector dimensions to account for eigenvector sign ambiguity. The graph-transformer input was

$$\mathbf{x}_{ip}^{(0)} = \mathbf{z}_{ip} + \mathbf{p}_p. \quad (12)$$

Pathway-aware graph transformer with soft masking

PATH uses $L = 2$ edge-aware graph transformer blocks with $H = 4$ attention heads. The pathway adjacency prior enters attention in two ways. First, non-adjacent pathway pairs receive a finite soft penalty rather than a hard mask,

$$m_{pq} = \begin{cases} 0, & A_{pq} > 0 \text{ or } p = q, \\ -10, & A_{pq} = 0 \text{ and } p \neq q. \end{cases} \quad (13)$$

This preserves a strong prior toward Reactome-connected pathways while allowing non-Reactome pathway pairs to retain small nonzero attention probability.

Second, the scalar edge value A_{pq} is used as an edge feature. At layer ℓ , it is projected to head-specific positive gains,

$$\phi_{pq}^{(\ell,h)} = \text{softplus}(w_h^{(\ell)} A_{pq} + b_h^{(\ell)}), \quad (14)$$

which are converted to an additive attention bias,

$$\phi_{pq}^{(\ell)} = \frac{1}{H} \sum_{h=1}^H \log(\phi_{pq}^{(\ell,h)} + \varepsilon). \quad (15)$$

The final attention bias is

$$M_{pq}^{(\ell)} = m_{pq} + \phi_{pq}^{(\ell)}. \quad (16)$$

For pathway-token matrix $\mathbf{X}_i^{(\ell-1)}$, multi-head attention is then computed as

$$\text{Attn}^{(\ell)}(\mathbf{X}) = \text{softmax} \left(\frac{\mathbf{Q}^{(\ell)} \mathbf{K}^{(\ell)\top}}{\sqrt{d/H}} + \mathbf{M}^{(\ell)} \right) \mathbf{V}^{(\ell)}. \quad (17)$$

Each transformer block applies residual connections, batch normalization, and a feed-forward network,

$$\hat{\mathbf{X}}^{(\ell)} = \text{BN} \left(\mathbf{X}^{(\ell-1)} + \text{Attn}^{(\ell)}(\mathbf{X}^{(\ell-1)}) \right), \quad (18)$$

$$\mathbf{X}^{(\ell)} = \text{BN} \left(\hat{\mathbf{X}}^{(\ell)} + \text{FFN}^{(\ell)}(\hat{\mathbf{X}}^{(\ell)}) \right). \quad (19)$$

The feed-forward network uses GELU activation[?], hidden dimension $4d$, and dropout 0.2. Edge features are also updated after each layer using a small MLP followed by batch normalization[?].

Patient-level readout and classification

After the final graph transformer layer, PATH pools pathway tokens into a patient representation using pathway-level attention,

$$\omega_{ip} = \frac{\exp\{\mathbf{a}^\top \tanh(\mathbf{W}_a \mathbf{x}_{ip}^{(L)})\}}{\sum_{q=1}^P \exp\{\mathbf{a}^\top \tanh(\mathbf{W}_a \mathbf{x}_{iq}^{(L)})\}}, \quad (20)$$

$$\mathbf{r}_i = \sum_{p=1}^P \omega_{ip} \mathbf{x}_{ip}^{(L)}. \quad (21)$$

The patient representation is passed through a classifier consisting of a linear layer, batch normalization, GELU activation, dropout, and a final two-class output layer,

$$\hat{\mathbf{y}}_i = f_{\text{cls}}(\mathbf{r}_i) \in \mathbb{R}^2. \quad (22)$$

Class probabilities are obtained with a softmax over the two logits.

4.3 Training and evaluation

Each main task was evaluated using stratified five-fold cross-validation. In each fold, 20% of samples were held out as the test set. The remaining 80% were split into training and validation subsets using stratified sampling, with 10% of the training pool used for validation. Thus, each fold used approximately 72% of samples for training, 8% for validation, and 20% for testing. Fold assignments were saved and reused for reproducibility.

The model was optimized using AdamW² with learning rate 10^{-4} , weight decay 5×10^{-4} , batch size 16, maximum 200 epochs, and dropout 0.2. Gradients were clipped to a maximum norm of 2.0. Early stopping monitored validation AUROC with patience 25 after a minimum of 50 epochs. The checkpoint with the highest validation AUROC was used for test evaluation.

Class imbalance was handled using inverse-frequency class weights. For class c , the weight was

$$w_c = \frac{N}{Cn_c}, \quad (23)$$

where N is the number of training samples, $C = 2$ is the number of classes, and n_c is the number of training samples in class c . The weighted cross-entropy loss was

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c \mathbb{I}(y_i = c) \log p_{ic}. \quad (24)$$

An optional focal-loss variant² was implemented but was not required for the main experiments unless explicitly stated.

For binary classification metrics, predicted class probabilities were converted to labels using the threshold that maximized F1 on the evaluated split:

$$\tau^* = \arg \max_{\tau} \frac{2 \text{Precision}(\tau) \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau) + 10^{-12}}. \quad (25)$$

Performance was quantified using AUROC, AUPRC, F1, precision, recall, and accuracy. Aggregate ROC and precision-recall curves were generated by interpolating fold-level curves onto a common grid and reporting mean curves with one-standard-deviation bands.

4.4 Interpretability analysis

PATH was interpreted at three biological scales: genes, pathways, and pathway-to-pathway interactions. Gene-level attribution was computed by applying SHAP GradientExplainer² to the trained model and summing mutation and CNV contributions for each gene,

$$\phi_{ig} = \phi_{ig}^{\text{mut}} + \phi_{ig}^{\text{cnv}}. \quad (26)$$

For a binary task, the class-associated gene score was

$$\Delta_g = \mathbb{E}[\phi_{ig} \mid y_i = 1] - \mathbb{E}[\phi_{ig} \mid y_i = 0], \quad (27)$$

where class 1 denotes the positive progression class, metastatic disease for the metastasis tasks and late stage for BLCA.

Pathway attribution was obtained by aggregating gene-level SHAP values over Reactome pathway membership,

$$\Phi_{ip} = \frac{1}{|\mathcal{G}_p|} \sum_{g \in \mathcal{G}_p} \phi_{ig}, \quad (28)$$

and pathways were ranked by

$$\Delta_p = \mathbb{E}[\Phi_{ip} \mid y_i = 1] - \mathbb{E}[\Phi_{ip} \mid y_i = 0]. \quad (29)$$

Top genes and pathways were cross-referenced with COSMIC, Reactome annotations, and prior literature.

Pathway-to-pathway rewiring was estimated from class-specific pathway attention matrices learned by the graph transformer. For each directed pathway pair (p, q) , the differential interaction score was

$$\Delta_{pq} = \bar{T}_{pq}^{(1)} - \bar{T}_{pq}^{(0)}, \quad (30)$$

where $\bar{T}_{pq}^{(1)}$ and $\bar{T}_{pq}^{(0)}$ are the mean attention-based interaction strengths in the positive and negative classes. Candidate non-Reactome edges were defined as pathway pairs with prior adjacency $A_{pq} = 0$ but nonzero learned interaction. Edge-level

comparisons were adjusted using the Benjamini-Hochberg false-discovery-rate procedure[?], and significant rewiring was defined at $q < 0.05$.

For hub analysis, pathway scores were combined with the prior pathway graph. Let $s_p = \max(\Delta_p, 0)$ be the positive class-associated pathway score. A SHAP-informed edge score was

$$E_{pq} = A_{pq}s_p s_q, \quad (31)$$

and the hub score for pathway p was

$$H_p = \sum_q E_{pq}. \quad (32)$$

This analysis prioritizes pathways that are both strongly attributed and connected to other positively attributed pathways.

4.5 Baseline models

PATH was compared against seven baseline models representing four major families of cancer prediction methods: pathway-informed neural networks, transformer-based pathway models, graph-based multi-omics models, and convolutional multi-omics models. The pathway-informed baselines were P-NET[?] and PathCNN[?]. The transformer-based pathway baseline was Pathformer[?]. The graph-based multi-omics baselines were MOGONET[?] and MOGAT[?]. The convolutional baselines were eiCNN and liCNN[?]. All baselines were evaluated using the same data splits, input modalities, and performance metrics as PATH.

4.6 Software and reproducibility

PATH was implemented in PyTorch. Data processing and evaluation used NumPy, pandas, scikit-learn, Matplotlib, and Seaborn. Random seeds were fixed for Python, NumPy, and PyTorch. Cross-validation splits, trained checkpoints, per-sample predictions, and pathway-importance outputs were written to disk for downstream analysis and reproducibility.

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Author contributions statement

Must include all authors, identified by initials, for example: A.A. conceived the experiment(s), A.A. and B.A. conducted the experiment(s), C.A. and D.A. analysed the results. All authors reviewed the manuscript.

Supplementary Tables

Table 1 Supplementary Table 1. All top-10 PANCANCER genes ranked by mean Δ SHAP ($\text{SHAP}_{\text{met}} - \text{SHAP}_{\text{prim}}$). Impact stages: Initiation = initiation; Progression = progression. $\Rightarrow$ denotes stepwise transition. Genes without direct metastasis evidence are flagged as hypothesis-generating.

Rank	Gene	Ref.	Impact stage	Transition chain
1	<i>BRAF</i>	?	Initiation $\rightarrow$ Progression	<i>BRAF</i> ^{V600E} mutation $\Rightarrow$ MEK-ERK activation $\Rightarrow$ cytoskeletal remodelling $\Rightarrow$ invasion
2	<i>TP53</i>	?	Initiation	<i>TP53</i> gain-of-function mutation $\Rightarrow$ pro-invasive transcription $\Rightarrow$ anoikis resistance $\Rightarrow$ metastasis
3	<i>GHR</i>	?	Progression	<i>GHR</i> overexpression $\Rightarrow$ JAK/STAT & PI3K-AKT activation $\Rightarrow$ pro-metastatic signalling $\Rightarrow$ metastatic spread
4	<i>ADAMTS18</i>	?	Progression	<i>ADAMTS18</i> silencing $\Rightarrow$ AKT/NF- κ B activation $\Rightarrow$ ECM remodelling $\Rightarrow$ invasion
5	<i>SCN11A</i>	—	—	<i>SCN11A</i> expression $\Rightarrow$ ion channel activity $\Rightarrow$ indirect cancer association (no direct metastasis evidence)
6	<i>XDH</i>	?	Progression	<i>XDH</i> loss $\Rightarrow$ TGF- β /Smad2-3 activation $\Rightarrow$ EMT induction $\Rightarrow$ pulmonary metastatic colonisation
7	<i>MITF</i>	?	Progression	<i>MITF</i> amplification $\Rightarrow$ lineage survival programme $\Rightarrow$ invasion competence $\Rightarrow$ metastatic outgrowth
8	<i>EP300</i>	?	Progression	<i>EP300</i> chromatin acetylation $\Rightarrow$ EMT gene activation $\Rightarrow$ E-cadherin repression $\Rightarrow$ invasion
9	<i>MUC16</i>	?	Progression	<i>MUC16</i> overexpression $\Rightarrow$ mesothelin binding $\Rightarrow$ peritoneal adhesion $\Rightarrow$ transcoelomic dissemination
10	<i>EEF1E1</i>	—	—	<i>EEF1E1</i> expression $\Rightarrow$ translational control $\Rightarrow$ indirect cancer association (no direct metastasis evidence)

Table 2 Supplementary Table 2. Per-cohort early-vs-late stage classification performance of PATH (CNV + gene expression) across all 11 TCGA cohorts. Values are mean $\pm$ s.d. over 5-fold stratified cross-validation on held-out test folds. N = cohort size; %L = fraction late-stage (III/IV). AUC and AUPR are threshold-free and are the primary metrics; accuracy, balanced accuracy (mean of per-class recall; robust to class imbalance), F1, precision and recall are reported at the per-fold F1-optimal operating point. Cohorts are ordered by test AUC; the horizontal rule separates the five cohorts clearing the interpretability gate ($\text{AUC} \geq 0.62$) used for driver analysis.

Cohort	N	%L	Acc	Bal. Acc	AUC	F1 _{bin}	F1 _{macro}	Prec	Rec
KIRP	259	26	0.838 $\pm$.05	0.778 $\pm$.02	0.808 $\pm$.05	0.684 $\pm$.05	0.787 $\pm$.04	0.760 $\pm$.18	0.656 $\pm$.08
KIRC	522	39	0.653 $\pm$.08	0.677 $\pm$.05	0.710 $\pm$.05	0.641 $\pm$.03	0.638 $\pm$.09	0.561 $\pm$.08	0.789 $\pm$.15
BLCA	402	68	0.712 $\pm$.03	0.570 $\pm$.05	0.669 $\pm$.08	0.820 $\pm$.01	0.536 $\pm$.09	0.711 $\pm$.03	0.971 $\pm$.03
COAD	430	43	0.567 $\pm$.10	0.605 $\pm$.08	0.647 $\pm$.05	0.646 $\pm$.04	0.521 $\pm$.15	0.512 $\pm$.06	0.898 $\pm$.10
LIHC	340	25	0.582 $\pm$.18	0.627 $\pm$.06	0.621 $\pm$.08	0.474 $\pm$.05	0.536 $\pm$.16	0.387 $\pm$.09	0.718 $\pm$.20
BRCA	1054	25	0.527 $\pm$.09	0.584 $\pm$.02	0.570 $\pm$.02	0.430 $\pm$.01	0.503 $\pm$.08	0.315 $\pm$.02	0.701 $\pm$.13
THCA	495	33	0.507 $\pm$.12	0.592 $\pm$.06	0.559 $\pm$.08	0.534 $\pm$.03	0.476 $\pm$.13	0.407 $\pm$.07	0.846 $\pm$.18
LUAD	504	21	0.466 $\pm$.14	0.577 $\pm$.03	0.544 $\pm$.04	0.386 $\pm$.02	0.441 $\pm$.10	0.274 $\pm$.06	0.770 $\pm$.18
HNSC	439	78	0.790 $\pm$.01	0.536 $\pm$.04	0.526 $\pm$.08	0.881 $\pm$.01	0.506 $\pm$.07	0.792 $\pm$.01	0.991 $\pm$.01
SKCM	333	49	0.526 $\pm$.02	0.533 $\pm$.02	0.508 $\pm$.08	0.671 $\pm$.01	0.408 $\pm$.05	0.510 $\pm$.01	0.982 $\pm$.02
STAD	388	54	0.562 $\pm$.01	0.523 $\pm$.01	0.507 $\pm$.05	0.711 $\pm$.01	0.402 $\pm$.03	0.553 $\pm$.01	0.995 $\pm$.01

Table 3 Supplementary Table 3. Stage-classification cohorts. Eleven TCGA cohorts used for early-vs-late stage prediction, each with matched copy-number variation (CNV) and gene-expression profiles over a shared set of 17,944 genes. Early = AJCC stage I/II; Late = stage III/IV. All cohorts were derived from the TCGA PanCanAtlas via UCSC Xena² and share the same Reactome pathway graph. Per-cohort performance is reported in Supplementary Table 2.

Cohort	Cancer type	N	Early (I/II)	Late (III/IV)	% Late	Modalities
KIRP	Kidney renal papillary cell carcinoma	259	192	67	26	CNV + Expr
KIRC	Kidney renal clear cell carcinoma	522	318	204	39	CNV + Expr
BLCA	Bladder urothelial carcinoma	402	130	272	68	CNV + Expr
COAD	Colon adenocarcinoma	430	243	187	43	CNV + Expr
LIHC	Liver hepatocellular carcinoma	340	255	85	25	CNV + Expr
BRCA	Breast invasive carcinoma	1054	786	268	25	CNV + Expr
THCA	Thyroid carcinoma	495	331	164	33	CNV + Expr
LUAD	Lung adenocarcinoma	504	396	108	21	CNV + Expr
HNSC	Head & neck squamous cell carcinoma	439	97	342	78	CNV + Expr
SKCM	Skin cutaneous melanoma	333	169	164	49	CNV + Expr
STAD	Stomach adenocarcinoma	388	178	210	54	CNV + Expr
Total	11 cohorts	5166	3095	2071	40	—

Table 4 Supplementary Table 4. Early-vs-late stage classification: PATH versus eight baselines on CNV + gene expression across all 11 TCGA cohorts. Values are mean $\pm$ s.d. test AUC over 5-fold stratified cross-validation, all models sharing the identical frozen train/validation/test splits, gene set and pathway graph as PATH (Supplementary Table 2). The best model per cohort is in **bold**. Cohorts are ordered by PATH test AUC; the rule separates the five cohorts clearing the interpretability gate ($\text{AUC} \geq 0.62$). Dashes denote runs not available.

Cohort	<i>N</i>	%L	PATH	CNN _{ei}	CNN _{li}	PathCNN	P-NET	Pathformer	DeePathNet	MOGONET	MOGAT
KIRP	259	26	0.808 $\pm$ 0.05	0.807 $\pm$ 0.08	0.794 $\pm$ 0.08	0.811 $\pm$ 0.04	0.824 $\pm$ 0.05	0.815 $\pm$ 0.06	0.806 $\pm$ 0.03	0.614 $\pm$ 0.15	0.789 $\pm$ 0.07
KIRC	522	39	0.710 $\pm$ 0.05	0.691 $\pm$ 0.04	0.705 $\pm$ 0.03	0.710 $\pm$ 0.05	0.768 $\pm$ 0.05	0.733 $\pm$ 0.04	0.738 $\pm$ 0.05	0.539 $\pm$ 0.12	0.739 $\pm$ 0.06
BLCA	402	68	0.669 $\pm$ 0.08	0.612 $\pm$ 0.04	0.608 $\pm$ 0.08	0.692 $\pm$ 0.07	0.704 $\pm$ 0.08	0.627 $\pm$ 0.04	0.685 $\pm$ 0.08	0.583 $\pm$ 0.13	0.619 $\pm$ 0.09
COAD	430	43	0.647 $\pm$ 0.05	0.560 $\pm$ 0.06	0.640 $\pm$ 0.07	0.644 $\pm$ 0.03	0.704 $\pm$ 0.05	0.650 $\pm$ 0.08	0.670 $\pm$ 0.04	0.506 $\pm$ 0.07	0.690 $\pm$ 0.09
LIHC	340	25	0.621 $\pm$ 0.08	0.513 $\pm$ 0.09	0.624 $\pm$ 0.07	0.546 $\pm$ 0.10	0.523 $\pm$ 0.04	0.547 $\pm$ 0.06	0.560 $\pm$ 0.06	0.482 $\pm$ 0.08	0.541 $\pm$ 0.04
BRCA	1054	25	0.570 $\pm$ 0.02	0.544 $\pm$ 0.04	0.560 $\pm$ 0.04	0.572 $\pm$ 0.02	0.604 $\pm$ 0.03	0.569 $\pm$ 0.03	0.604 $\pm$ 0.02	0.482 $\pm$ 0.03	0.576 $\pm$ 0.05
THCA	495	33	0.559 $\pm$ 0.08	0.571 $\pm$ 0.03	0.552 $\pm$ 0.02	0.654 $\pm$ 0.05	0.686 $\pm$ 0.04	0.630 $\pm$ 0.05	0.673 $\pm$ 0.05	0.500 $\pm$ 0.07	0.638 $\pm$ 0.06
LUAD	504	21	0.544 $\pm$ 0.04	0.501 $\pm$ 0.08	0.542 $\pm$ 0.06	0.584 $\pm$ 0.07	0.579 $\pm$ 0.04	0.568 $\pm$ 0.09	0.626 $\pm$ 0.07	0.472 $\pm$ 0.06	0.612 $\pm$ 0.04
HNSC	439	78	0.526 $\pm$ 0.08	0.601 $\pm$ 0.10	0.623 $\pm$ 0.06	0.544 $\pm$ 0.08	0.627 $\pm$ 0.10	0.514 $\pm$ 0.04	0.569 $\pm$ 0.04	0.450 $\pm$ 0.11	0.511 $\pm$ 0.04
SKCM	333	49	0.508 $\pm$ 0.08	0.452 $\pm$ 0.07	0.461 $\pm$ 0.07	0.543 $\pm$ 0.09	0.573 $\pm$ 0.06	0.498 $\pm$ 0.03	0.533 $\pm$ 0.07	0.522 $\pm$ 0.06	0.547 $\pm$ 0.04
STAD	388	54	0.507 $\pm$ 0.05	0.539 $\pm$ 0.08	0.510 $\pm$ 0.02	0.564 $\pm$ 0.03	0.597 $\pm$ 0.05	0.523 $\pm$ 0.05	0.563 $\pm$ 0.08	0.466 $\pm$ 0.07	0.571 $\pm$ 0.03

Table 5 Supplementary Table 5. Leave-one-cohort-out (LOCO) early-vs-late stage classification: PATH versus eight baselines on CNV + gene expression. Each model is trained on the pooled remaining ten cohorts and tested on the entire held-out cohort (single fold); values are test AUC, best model per cohort in **bold**. All models share the identical pooled training population, gene set and pathway graph. Dashes denote runs not available.

Cohort	PATH	CNN _{ei}	CNN _{li}	PathCNN	P-NET	Pathformer	DeePathNet	MOGONET	MOGAT
KIRP	0.770	0.677	0.605	0.702	0.587	0.603	0.759	0.398	0.797
KIRC	0.628	0.631	0.622	0.590	0.720	–	0.670	0.596	0.668
BLCA	0.584	0.577	0.541	0.573	0.563	0.620	0.552	0.538	0.627
COAD	0.562	0.548	0.594	0.571	0.625	–	0.579	0.364	0.572
LIHC	0.595	0.607	0.561	0.606	0.516	0.467	0.483	0.522	0.525
BRCA	0.526	0.528	0.530	0.484	0.506	0.518	0.513	0.499	0.525
THCA	0.555	0.517	0.593	0.549	0.580	0.620	0.538	0.455	0.593
LUAD	0.622	0.544	0.573	0.555	0.594	0.575	0.548	0.474	0.578
HNSC	0.526	0.577	0.459	0.520	0.548	0.569	0.535	0.527	0.569
SKCM	0.508	0.515	0.546	0.512	0.553	0.498	0.482	0.478	0.510
STAD	0.502	0.471	0.527	0.451	0.478	0.504	0.478	0.514	0.510

Table 6 Supplementary Table 6. Leave-one-cohort-out (LOCO) early-vs-late stage classification on CNV + mutation: full evaluation metrics for PATH and eight baselines across twelve TCGA cohorts. Each model is trained on the pooled remaining eleven cohorts and tested on the entire held-out cohort (single fold), under an identical protocol (train-only CNV *z*-scoring, cohort-balanced 10% validation split, F_1 -optimal decision threshold). Best value per row is in **bold**; per-metric means are italicised. All models remain near chance on every metric, indicating that CNV + mutation carries no transferable cross-cancer stage signal.

Metric	Cohort	PATH	CNN _{ei}	CNN _{li}	PathCNN	P-NET	Pathformer	DeePathNet	MOGONET	MOGAT
AUC-ROC	BLCA	0.475	0.526	0.476	0.489	0.468	0.501	0.505	0.515	0.562
	BRCA	0.509	0.502	0.480	0.524	0.518	0.550	0.491	0.500	0.535
	COAD	0.602	0.407	0.577	0.620	0.573	0.497	0.597	0.562	0.529
	HNSC	0.536	0.510	0.505	0.480	0.506	0.419	0.416	0.477	0.487
	KIRC	0.450	0.541	0.510	0.478	0.493	0.510	0.453	0.514	0.516
	KIRP	0.388	0.406	0.403	0.414	0.311	0.447	0.389	0.356	0.438
	LIHC	0.435	0.444	0.461	0.496	0.468	0.462	0.505	0.500	0.443
	LUAD	0.511	0.480	0.512	0.541	0.510	0.470	0.486	0.436	0.537
	LUSC	0.527	0.482	0.527	0.518	0.493	0.434	0.487	0.542	0.507
	SKCM	0.517	0.556	0.524	0.536	0.539	0.500	0.536	0.494	0.546
	STAD	0.518	0.508	0.534	0.537	0.525	0.471	0.553	0.458	0.532
	THCA	0.459	0.444	0.523	0.511	0.436	0.549	0.484	0.554	0.495
Mean		0.494	0.484	0.503	0.512	0.487	0.484	0.492	0.492	0.511
Bal. Acc.	BLCA	0.508	0.500	0.500	0.508	0.500	0.504	0.500	0.504	0.517
	BRCA	0.511	0.501	0.506	0.503	0.501	0.519	0.500	0.505	0.500
	COAD	0.604	0.503	0.506	0.592	0.510	0.500	0.500	0.510	0.500
	HNSC	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
	KIRC	0.505	0.552	0.546	0.554	0.500	0.505	0.515	0.518	0.528
	KIRP	0.503	0.500	0.505	0.500	0.500	0.514	0.500	0.500	0.500
	LIHC	0.506	0.500	0.500	0.560	0.528	0.502	0.500	0.512	0.510
	LUAD	0.516	0.527	0.527	0.563	0.524	0.502	0.508	0.503	0.521
	LUSC	0.512	0.523	0.546	0.512	0.505	0.510	0.516	0.566	0.500
	SKCM	0.515	0.515	0.500	0.500	0.513	0.523	0.502	0.502	0.520
	STAD	0.500	0.500	0.514	0.547	0.500	0.503	0.500	0.521	0.500
	THCA	0.500	0.519	0.503	0.500	0.500	0.500	0.500	0.500	0.500
Mean		0.515	0.512	0.513	0.528	0.507	0.507	0.503	0.512	0.508
AUPR	BLCA	0.660	0.706	0.666	0.666	0.657	0.681	0.693	0.691	0.721
	BRCA	0.232	0.227	0.221	0.241	0.245	0.262	0.227	0.242	0.246
	COAD	0.562	0.375	0.525	0.523	0.533	0.479	0.546	0.536	0.457
	HNSC	0.823	0.803	0.786	0.778	0.804	0.735	0.743	0.765	0.780
	KIRC	0.359	0.390	0.399	0.352	0.395	0.418	0.373	0.379	0.395
	KIRP	0.229	0.223	0.226	0.212	0.194	0.219	0.237	0.212	0.253
	LIHC	0.233	0.237	0.270	0.258	0.273	0.243	0.295	0.293	0.247
	LUAD	0.244	0.220	0.237	0.233	0.219	0.207	0.242	0.208	0.247
	LUSC	0.218	0.189	0.207	0.198	0.198	0.165	0.196	0.203	0.193
	SKCM	0.457	0.478	0.483	0.465	0.478	0.437	0.475	0.434	0.486
	STAD	0.567	0.554	0.581	0.566	0.570	0.531	0.588	0.521	0.565
	THCA	0.325	0.313	0.405	0.380	0.317	0.368	0.332	0.386	0.330
Mean		0.409	0.393	0.417	0.406	0.407	0.395	0.412	0.406	0.410
Macro-F ₁	BLCA	0.810	0.808	0.808	0.810	0.808	0.809	0.808	0.809	0.812
	BRCA	0.375	0.374	0.376	0.375	0.374	0.382	0.374	0.376	0.374
	COAD	0.614	0.615	0.615	0.642	0.619	0.614	0.614	0.619	0.614
	HNSC	0.875	0.875	0.875	0.875	0.875	0.875	0.875	0.875	0.875
	KIRC	0.545	0.564	0.562	0.569	0.543	0.545	0.545	0.552	0.556
	KIRP	0.403	0.401	0.404	0.401	0.401	0.408	0.401	0.401	0.401
	LIHC	0.406	0.403	0.403	0.433	0.413	0.404	0.403	0.409	0.408
	LUAD	0.364	0.369	0.368	0.383	0.364	0.358	0.361	0.359	0.365
	LUSC	0.326	0.325	0.329	0.326	0.321	0.324	0.322	0.343	0.321
	SKCM	0.623	0.621	0.618	0.618	0.620	0.625	0.619	0.619	0.625
	STAD	0.707	0.707	0.712	0.710	0.707	0.708	0.707	0.711	0.707
	THCA	0.498	0.506	0.499	0.498	0.498	0.498	0.498	0.498	0.498
Mean		0.545	0.547	0.548	0.553	0.545	0.546	0.544	0.547	0.546

Table 7 Supplementary Table 7. Intra-cohort early-vs-late stage classification: PATH versus eight baselines on CNV + mutation (genuine MC3 somatic mutation) across the eight TCGA cohorts with >400 samples carrying both modalities. Values are mean $\pm$ s.d. test AUC over 5-fold stratified cross-validation on the identical frozen splits, gene set and pathway graph. Best model per cohort in **bold**. Cohorts ordered by PATH test AUC.

Cohort	N	%L	PATH	CNN _{ci}	CNN _{li}	PathCNN	P-NET	Pathformer	DeePathNet	MOGONET	MOGAT
KIRP	247	25	0.761±0.09	0.741±0.10	0.743±0.11	0.777±0.08	0.784±0.06	0.780±0.11	0.752±0.09	0.549±0.10	0.786 ±0.04
KIRC	349	37	0.595±0.04	0.615±0.03	0.534±0.04	0.620±0.05	0.624±0.06	0.594±0.07	0.674 ±0.06	0.505±0.06	0.662±0.04
THCA	480	33	0.587±0.05	0.575±0.02	0.577±0.03	0.575±0.07	0.629±0.07	0.574±0.05	0.589±0.04	0.481±0.11	0.679 ±0.02
BRCA	766	23	0.580±0.06	0.503±0.02	0.496±0.05	0.563±0.07	0.610 ±0.07	0.512±0.05	0.575±0.03	0.471±0.03	0.541±0.03
COAD	262	44	0.596±0.07	0.659±0.06	0.642±0.06	0.679 ±0.04	0.678±0.04	0.667±0.05	0.622±0.08	0.466±0.10	0.637±0.05
LUAD	487	22	0.536±0.09	0.546±0.02	0.542±0.04	0.582±0.07	0.613 ±0.07	0.544±0.08	0.581±0.09	0.509±0.07	0.476±0.07
HNSC	428	78	0.533±0.11	0.583±0.10	0.600 ±0.11	0.599±0.09	0.572±0.07	0.555±0.03	0.573±0.06	0.479±0.08	0.546±0.07
LUSC	460	19	0.520±0.10	0.541±0.01	0.481±0.06	0.580±0.04	0.581 ±0.04	0.533±0.11	0.576±0.05	0.536±0.11	0.548±0.06
LIHC	333	25	0.492±0.10	0.517±0.03	0.552±0.07	0.576 ±0.06	0.538±0.06	0.537±0.08	0.555±0.07	0.519±0.08	0.502±0.02
SKCM	427	45	0.497±0.08	0.516±0.02	0.555±0.04	0.585 ±0.04	0.536±0.07	0.538±0.06	0.513±0.03	0.531±0.03	0.492±0.03
BLCA	403	68	0.463±0.05	0.508±0.04	0.438±0.06	0.478±0.06	0.521±0.03	0.481±0.05	0.497±0.03	0.533 ±0.09	0.500±0.07
STAD	408	55	0.440±0.06	0.500±0.06	0.502±0.03	0.508±0.10	0.497±0.07	0.475±0.06	0.548 ±0.12	0.534±0.09	0.469±0.03

Pathway interaction network (Fig. 6b). Nodes are coloured by their *Reactome top-level category*. We obtain each group by traversing the Reactome pathway hierarchy (the `ReactomePathwaysRelation` table) from a pathway up to its root ancestor, so the grouping is taken directly from Reactome rather than assigned by hand² (Supplementary Table 8). The network's *novel* interactions are pathway pairs that PATH assigns high learned attention but that are *absent* from the Reactome interaction prior enforced by the soft-mask. We deem a novel interaction valid when its attention differs significantly between metastatic and primary tumours after Benjamini–Hochberg false-discovery-rate control ($q < 0.05$). Three of the eight top candidates pass this test (Supplementary Table 9); each links two *distinct* Reactome top-level categories—extracellular matrix organization↔circadian clock, and cell/cell–cell junction organization↔cilium assembly—so PATH proposes crosstalk between programmes that are not connected in the curated Reactome graph.

Table 8 Supplementary Table 8. Reactome top-level grouping of the pathways in the interaction network (Fig. 6b). Each pathway's group is its Reactome top-level (root) category, obtained by traversing the Reactome pathway hierarchy (`ReactomePathwaysRelation`) from the pathway to its root ancestor². *Circadian Clock* is itself a Reactome top-level pathway. Node colours in Fig. 6b encode these groups.

Pathway	Reactome ID	Reactome top-level group
Adherens junctions interactions	R-HSA-418990	Cell-Cell communication
Cell junction organization	R-HSA-446728	Cell-Cell communication
Cell-Cell communication	R-HSA-1500931	Cell-Cell communication
Cell-cell junction organization	R-HSA-421270	Cell-Cell communication
Circadian Clock	R-HSA-400253	Circadian Clock
Extracellular matrix organization	R-HSA-1474244	Extracellular matrix organization
Cardiac conduction	R-HSA-5576891	Muscle contraction
Muscle contraction	R-HSA-397014	Muscle contraction
Cilium Assembly	R-HSA-5617833	Organelle biogenesis and maintenance
Activated PKN1 stimulates transcription of AR (androgen receptor) regulated genes KLK2 and KLK3	R-HSA-5625886	Signal Transduction
RHO GTPases activate PKNs	R-HSA-5625740	Signal Transduction

Table 9 Supplementary Table 9. All PATH-inferred novel pathway–pathway interactions (Fig. 6b). Candidate novel interactions are pathway pairs that receive high learned attention in PATH but are *absent* from the Reactome interaction prior (soft-mask). For each candidate we test whether its attention differs between metastatic and primary tumours and control the false discovery rate with the Benjamini–Hochberg procedure; interactions with $q < 0.05$ (✓) are considered validated. Learned weight is the mean attention assigned by PATH; Δ is metastatic–primary attention.

Pathway A	Pathway B	Learned wt.	Δ (met–prim)	q -value	Valid
Extracellular matrix organization	Circadian Clock	0.225	+0.0129	4.8e-08	✓
Cell junction organization	Cilium Assembly	0.115	-0.0098	5.6e-05	✓
Cell-cell junction organization	Cilium Assembly	0.113	-0.0102	2.4e-05	✓
Cell junction organization	Digestion	0.090	+0.0006	2.6e-01	–
Cell-cell junction organization	Digestion	0.090	+0.0004	5.2e-01	–
Cell-Cell communication	Digestion	0.088	+0.0003	6.4e-01	–
Cell junction organization	Digestion and absorption	0.086	-0.0008	2.0e-01	–
Cell-cell junction organization	Digestion and absorption	0.086	-0.0009	1.2e-01	–

Table 10 Summary of PATH-derived cancer progression signals. The table summarizes gene-level, pathway-level, and pathway–pathway interaction signals identified by PATH and their support from prior literature or existing pathway databases.

Alteration class	Evidence status	Pan-cancer	Prostate	BLCA
Gene	Literature-confirmed	9	8	7
	Not confirmed	1	2	3
Pathway	Literature-confirmed	7	8	6
	Not/partially confirmed	3	2	4
Pathway–pathway interaction	Present in database	7	8	5
	Literature-confirmed	1	1	3
	Not confirmed	2	1	2

Note: BLCA, bladder urothelial carcinoma.

Table 11 (a) We confirmed important genes reported by PATH (see Column **Ref**). Impact stages follow ref. [1]. $\Rightarrow$ denotes stepwise transition.

Dataset	Gene	Ref.	Impact stage [1]	Transition chain
PAN-CANCER	<i>BRAF</i>	?	Initiation $\rightarrow$ Progression	<i>BRAF</i> ^{V600E} mutation $\Rightarrow$ MEK–ERK activation $\Rightarrow$ cytoskeletal remodelling $\Rightarrow$ invasion
	<i>TP53</i>	?	Initiation	TP53 gain-of-function mutation $\Rightarrow$ pro-invasive transcription $\Rightarrow$ anoikis resistance $\Rightarrow$ metastasis
	<i>GHR</i>	?	Progression	GHR overexpression $\Rightarrow$ JAK/STAT & PI3K–AKT activation $\Rightarrow$ pro-metastatic signalling $\Rightarrow$ metastatic spread
PRAD	<i>AR</i>	?	Initiation $\rightarrow$ Progression	AR mutation & amplification $\Rightarrow$ androgen-independent transcription $\Rightarrow$ castration resistance $\Rightarrow$ metastasis
	<i>AKT1</i>	?	Initiation $\rightarrow$ Progression	AKT1 activation $\Rightarrow$ PI3K–AKT signalling $\Rightarrow$ cell motility & survival $\Rightarrow$ metastasis
	<i>PTEN</i>	?	Initiation	PTEN loss $\Rightarrow$ PI3K–AKT amplification $\Rightarrow$ mTOR activation $\Rightarrow$ castration resistance
BLCA	<i>PTPN11</i>	?,?	Initiation $\rightarrow$ Progression	PTPN11/SHP2 activation $\Rightarrow$ RTK-to-RAS signalling $\Rightarrow$ MAPK output $\Rightarrow$ late-stage BLCA programme
	<i>PSMB5</i>	?,?	Progression $\rightarrow$ Establishment	Proteasome subunit activity $\Rightarrow$ protein-turnover remodelling $\Rightarrow$ tumour-cell survival $\Rightarrow$ progression
	<i>KRAS</i>	?,?,?	Initiation $\rightarrow$ Progression	KRAS alteration $\Rightarrow$ RAS–MAPK signalling $\Rightarrow$ proliferation and survival $\Rightarrow$ late-stage transition

Table 12 (b) Pathway-level alterations **Update the caption see Table 2(a).** Columns as in (a).

Dataset	Pathway	Ref.	Impact stage [1]	Mechanism chain
PAN-CANCER	<i>Sprouty</i> reg. of <i>FGF</i> signaling	?	Progression	FGF feedback suppression loss $\Rightarrow$ FGFR sustained activation $\Rightarrow$ invasion signalling $\Rightarrow$ metastasis
	RAS–RAF–MEK–ERK cascade	?	Progression $\rightarrow$ Establishment	RAF/BRAF activation $\Rightarrow$ MEK/ERK phosphorylation $\Rightarrow$ invasion & survival $\Rightarrow$ metastatic spread
PRAD	PKN1/AR target gene transcription	?,?,?	Initiation $\rightarrow$ Progression	PKN1 activation $\Rightarrow$ AR transcription of <i>KLK2/KLK3</i> $\Rightarrow$ androgen-independent signalling $\Rightarrow$ castration resistance
	SUMOylation of intracellular receptors	?,?,?	Initiation $\rightarrow$ Progression	AR SUMOylation $\Rightarrow$ selective receptor target-gene regulation $\Rightarrow$ steroid-receptor rewiring $\Rightarrow$ castration resistance
	RUNX2 osteoblast differentiation	?,?,?	Progression	RUNX2 activation $\Rightarrow$ osteoblast programme hijacking $\Rightarrow$ bone microenvironment remodelling $\Rightarrow$ bone metastasis
BLCA	FRS-mediated FGFR3 signalling	?,?,?,?	Initiation $\rightarrow$ Progression	FGFR3 activation $\Rightarrow$ FRS2 adaptor recruitment $\Rightarrow$ RAS/MAPK signalling $\Rightarrow$ BLCA stage signal
	FRS-mediated FGFR1 signalling	?,?,?	Initiation $\rightarrow$ Progression	FGFR1 activation $\Rightarrow$ FRS2 coupling $\Rightarrow$ MAPK/PI3K output $\Rightarrow$ growth-factor programme
	Downstream signalling of activated FGFR3	?,?,?,?	Progression	Activated FGFR3 $\Rightarrow$ downstream kinase signalling $\Rightarrow$ urothelial carcinoma growth $\Rightarrow$ stage classification

Table 13 (c) **Update the caption see table 2: Pathway–pathway rewiring alterations.** Columns as in (a).

Dataset	Interaction	Ref.	Impact stage [1]	Rewiring chain
PRAD PAN-CANCER	PI3K ↔ MAPK	[20]	Initiation → Progression	PI3K/MAPK co-activation ⇒ bypass of single-pathway inhibition ⇒ therapy resistance
	p53 ↔ Rb	[21]	Initiation	p53 loss ⇒ Rb deregulation ⇒ dual checkpoint failure ⇒ genomic instability
	AR ↔ PI3K	[22]	Progression	AR inhibition ⇒ PI3K upregulation ⇒ compensatory survival ⇒ castration resistance
	mTOR ↔ AR	[23]	Progression → Establishment	mTOR activation ⇒ AR splice variant induction ⇒ ligand-independent growth ⇒ metastasis
BLCA	FGFR ↔ PI3K	[24]	Initiation → Progression	FGFR3 mut. ⇒ PI3K crosstalk ⇒ dual proliferative signalling ⇒ tumour progression
	Rb ↔ p53	[25]	Progression → Establishment	Rb/p53 co-loss ⇒ checkpoint collapse ⇒ aggressive invasion ⇒ muscle-invasive BLCA